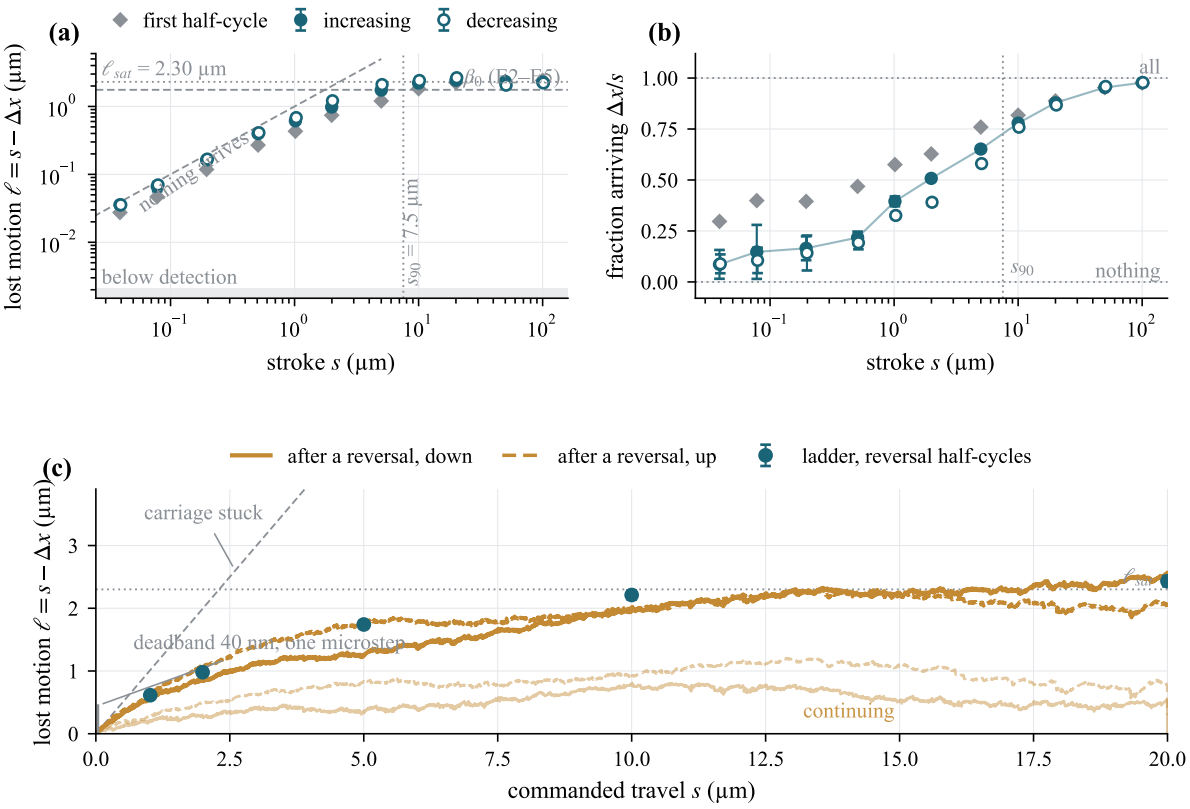

(a)
first half-cycle
increasing
decreasing
ℓ_{sat} = 2.30 µm
β_0 (B2–B5)
nothing arrives
below detection
s_{90} = 7.5 µm
lost motion $\ell = s - \Delta x$ (µm)
stroke s (µm)
(b)
all
nothing
s_{90}
fraction arriving $\Delta x/s$
stroke s (µm)
(c)
after a reversal, down
after a reversal, up
ladder, reversal half-cycles
carriage stuck
deadband 40 nm, one microstep
continuing
lost motion $\ell = s - \Delta x$ (µm)
commanded travel s (µm)